

∴ All three inscriptions cannot be True, and the Queen cannot say this.

19) Each inhabitant of a remote village always tells the truth or always lies. A villager will give only a "Yes" or a "No" response to a question a tourist asks. Suppose you are a tourist visiting this area and come to a fork in the road. One branch leads to the ruins you want to visit; the other branch leads deep into the jungle. A villager is standing at the fork in the road. What one question can you ask the villager to determine which branch take? (Famously difficult)
(Very deep question. Needs further study)

Soln:

Suppose, I ask the question: "Does the left branch lead to the ruins?"
The villager responds Yes. I have no way of determining whether the villager is speaking the Truth or not. A similar situation arises if the villager responds No.

I will try to create a question such that if q : "The left branch lead to the ruins" is True, the answer will be Yes, else if it is False, the answer will be No. q will be a part of the proposition.

The question asked will be: "You always speak the truth if and only if the left branch leads to the ruins". $p \rightarrow$ You always speak the truth.

I claim that this question works.

Proof:

	p	q	A	Output
①	T	T	Y	T
②	T	F	N	F
③	F	T	Y	F
④	F	F	N	T

When the villager is a truth teller and the left branch leads to the ruins, obviously the answer will be Yes. (Case 1)

When the villager is a truth teller but the left branch does not lead to the ruins, the question's truth value is False. The villager should truthfully say No (Case 2)

When the villager is a liar but the left branch leads to the ruins, the question's truth value is False. The villager lies and says Yes. (Case 3)
When the villager is a liar and the left branch does not lead to the ruins, the question's truth value is True. The villager